



# Multi-Decadal Assessment of Riverbank Dynamics and Shoreline Change Along the Sombreiro River, Southern Nigeria

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**Abstract:** Multi-decadal riverbank dynamics along the Sombreiro River in the Niger Delta, southern Nigeria, were quantified using four single-date Landsat images acquired in 1980, 2000, 2010, and 2025, integrated with geographic information system (GIS) analysis and the Digital Shoreline Analysis System (DSAS). The assessment covered the mapped Bukuma–Tombia–Buguma reach, comprising approximately 68.85 km of river channel and 137.70 km of bankline. Historical banklines were delineated from the image-derived land–water boundary and evaluated using the end point rate (EPR), linear regression rate (LRR), weighted linear regression rate (WLR), net shoreline movement (NSM), and shoreline change envelope (SCE). The EPR, LRR, and WLR were treated as rate-of-change statistics, whereas NSM and SCE were treated as displacement statistics. Mean rates of  $-1.63$ ,  $-1.37$ , and  $-1.48$  m yr<sup>-1</sup> were obtained for the EPR, LRR, and WLR, respectively. The NSM and SCE values originally reported as rates correspond to approximate displacements of  $-63.90$  and  $76.95$  m, respectively, when converted over the 45-year observation interval; however, these values require verification against the original DSAS output. Spatial analysis indicated that 46.98 km (34.0%) of the mapped bankline experienced erosion, 32.45 km (23.6%) experienced accretion, and 58.27 km (42.4%) remained stable. Given the four observations and uneven temporal intervals, the metrics were interpreted as multi-decadal changes and temporal trends rather than continuous annual migration. Potential positional variability associated with acquisition season, river stage, and tidal conditions was considered using a conservative half-pixel screening threshold of approximately 33.5 m. Overall, spatially heterogeneous riverbank behavior was identified, with net retreat exceeding accretion, demonstrating both the utility and limitations of multi-temporal Landsat imagery and DSAS for reconstructing long-term riverbank dynamics in tidally influenced environments.

**Keywords:** Riverbank erosion; Shoreline change; Sombreiro River; Digital Shoreline Analysis System; Remote sensing; Geographic information system; Niger Delta; Landsat; River morphology

## 1 Introduction

Riverbank erosion, bank retreat and accretion are natural geomorphic processes that continuously modify river channels, floodplains and adjacent environments [1–3]. Their magnitude and spatial expression are influenced by flow conditions, channel geometry, sediment supply and bank material, while human interventions can modify sediment transport and channel adjustment [3, 4]. In particular, sand extraction can alter channel and bank conditions [5], while navigation, dredging, vegetation clearance, settlement expansion and other forms of development may increase local pressure on unstable banks [4]. The consequences of bank erosion are not only geomorphic; they can also affect riparian communities, infrastructure and livelihoods [6]. The Niger Delta is among the world's major tropical deltaic environments and contains a dense network of distributary rivers, tidal creeks, wetlands and mangrove systems that support transportation, fisheries, agriculture, tourism and other river-dependent activities [7, 8]. Its hydrological setting and channel morphology are dynamic, with regional research emphasizing the importance of hydrological processes and cross-sectional adjustment in controlling channel change [9, 10]. Sediment transport is also central

to delta evolution, and recent evidence indicates substantial changes in suspended-sediment conditions in global river deltas [11]. These characteristics make distributary channels in the Niger Delta susceptible to continuing morphological adjustment under both natural and human pressures [4].

The Sombreiro River is an important distributary system in Rivers State and supports fisheries, navigation and other river-dependent activities. The basin is environmentally dynamic, but the present study deliberately narrows the quantitative analysis to the mapped Bukuma–Tombia–Buguma reach. Accordingly, the study area in this research refers to that analyzed reach rather than the full hydrological basin. Understanding historical bank positions is important for estimating erosion and accretion rates and for developing appropriate river-management interventions. Conventional field surveys can provide detailed observations but are difficult to apply repeatedly over long river reaches and long time periods. Satellite remote sensing and geographic information system (GIS) techniques provide a practical means of reconstructing historical bank positions and examining spatially distributed channel change. Recent studies have demonstrated the application of multi-temporal optical and radar imagery to quantify riverbank erosion, accretion and channel morphology [12–17].

A widely used framework for analyzing temporal shoreline and riverbank movement is the Digital Shoreline Analysis System (DSAS), which generates transects from a baseline and calculates shoreline-change statistics from multiple shoreline positions [18]. The present study uses the end point rate (EPR), linear regression rate (LRR), weighted linear regression rate (WLR), net shoreline movement (NSM) and shoreline change envelope (SCE), following the definitions and units in the DSAS documentation [18]. The literature demonstrates that riverbank change is spatially heterogeneous and may vary substantially through time. Studies using DSAS and other geospatial approaches have quantified erosion and accretion along the Ganga and other river systems [12, 13], while Barak River investigations have documented changes in channel morphology and centerline migration [14] as well as spatiotemporal inconsistency in erosion and deposition [15]. Other recent studies have examined riverbank dynamics using time-series radar imagery [13], geospatial analysis based on the autoregressive integrated moving average [16], and the implications of bank erosion for local livelihoods [17]. Together, these studies show the value of combining spatial analysis with temporal observations when interpreting riverbank adjustment.

The physical interpretation of bank movement also benefits from established fluvial-process concepts. Channel curvature, flow distribution and meander behavior can produce alternating zones of bank erosion and deposition [3, 19], while studies of incised channels emphasize the interaction among channel processes, bank erosion and management responses [20]. Approaches to defining erodible river corridors further demonstrate the importance of identifying bank-instability zones when planning sustainable river management [21]. These perspectives provide a broader process and management context for interpreting the spatial patterns identified in the Sombreiro River. The present study extends this body of work to the Sombreiro River by combining five shoreline-change metrics derived using DSAS with spatial erosion/accretion classification and cross-sectional analysis over a 45-year period. This approach provides a complementary assessment of long-term riverbank dynamics in the Niger Delta, where spatially variable erosion and associated channel-width changes are of particular management concern.

The lack of consistent long-term information on bank movement in many Niger Delta distributaries limits evidence-based management. Remote sensing provides a practical means of reconstructing historical bank positions, but the reliability of the result depends on image resolution, co-registration, extraction rules, acquisition conditions and temporal sampling [12, 13, 18]. The present study therefore evaluates both the observed long-term change and the uncertainty associated with the four-image record.

The four Landsat observations from 1980, 2000, 2010 and 2025 were analyzed with the GIS and DSAS. Because the intervals are uneven and each epoch is represented by a single image, the study estimates multi-decadal displacement and fitted trends rather than continuous annual bank migration. Acquisition-season, water-level and tidal-stage differences are recognized as potential contributors to the measured positional variation.

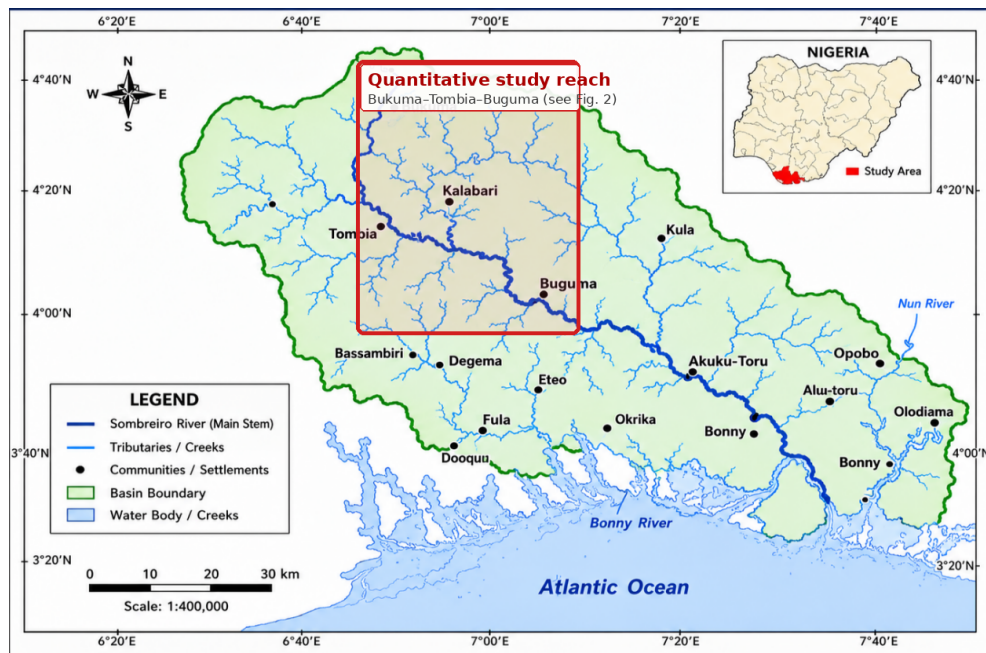
The principal contribution of the study is a reach-specific, multi-decadal assessment that combines DSAS statistics, spatial bankline classification and representative cross-sectional interpretation. The revised manuscript explicitly defines the mapped extent, distinguishes rate and distance statistics, describes the bankline extraction protocol, and treats positional uncertainty and sparse temporal sampling as central limitations. The resulting evidence is intended to support targeted monitoring and management of erosion-prone river corridors [20, 21].

## 2 Materials and Methods

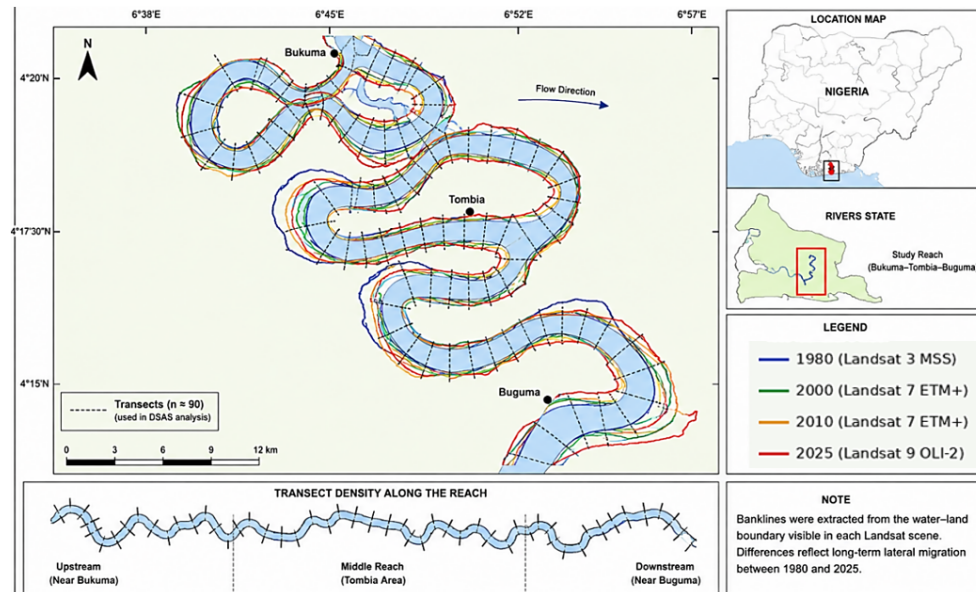
### 2.1 Study Area

The Sombreiro River Basin is situated in Rivers State in the eastern Niger Delta of southern Nigeria. The basin shown in Figure 1 is presented only for regional context and includes a broad network of distributary channels, tidal creeks and settlements. The names Bukuma, Tombia, Buguma, Degema, Abonnema, Bonny, Opobo and other named places on Figure 1 are communities/settlements, not separate rivers. The present quantitative analysis does not cover the entire basin. It is restricted to the mapped Sombreiro River reach extending from the Bukuma upstream endpoint through Tombia to the Buguma downstream endpoint. The mapped quantitative extent is approximately

6°38'–6°57'E and 4°15'–4°20'N. The mapped bankline length is 137.70 km for both banks combined, corresponding to an approximate 68.85 km channel reach. Accordingly, all erosion, accretion, stability, and shoreline-change results derived using DSAS refer to this defined reach and should not be extrapolated to the lower estuarine basin beyond the spatial extent shown in Figure 2.



**Figure 1.** Location map of the Sombreiro River Basin, Southern Nigeria



**Figure 2.** Mapped banklines of the Sombreiro River (Bukuma–Tombia–Buguma reach): 1980, 2000, 2010 and 2025

Note: DSAS = Digital Shoreline Analysis System; MSS = Multispectral Scanner System; ETM+ = Enhanced Thematic Mapper Plus; OLI-2 = Operational Land Imager-2.

The Sombreiro River is an important distributary waterway within the Niger Delta and supports navigation, fisheries, agriculture and other livelihood activities. Because the present analysis is confined to the Bukuma–Tombia–Buguma reach, statements concerning estuarine or lower-coast processes are treated as contextual explanations rather than as direct observations from portions of the basin that are outside the mapped quantitative extent.

The regional climate is humid tropical and has a pronounced seasonal cycle:

- Annual rainfall is approximately 2,500–4,500 mm.
- Mean annual temperature is approximately 26–28 °C.
- Relative humidity is generally high.
- The relatively drier period is commonly November–February.
- The wetter period generally extends from March to October.

The geology is dominated by recent alluvial and deltaic deposits, including sands, silts, clays, peat and organic-rich swamp sediments. Such materials can be susceptible to bank erosion and mass failure, particularly where vegetative protection is disturbed.

The Sombreiro reach lies within the low-lying, sediment-rich Niger Delta, where the channel is developed largely in recent alluvial and deltaic deposits. The bank materials identified in the regional setting include unconsolidated sands, silts and clays, with peat and organic-rich swamp sediments occurring in wetland environments. These relatively weak and heterogeneous materials provide a plausible physical basis for bank retreat, local mass failure and rapid adjustment where vegetative protection is reduced. Geomorphologically, the mapped Bukuma–Tombia–Buguma reach is a distributary channel within a broad deltaic floodplain rather than a confined bedrock valley. The absence of strong bedrock confinement permits lateral channel adjustment and meander migration, while contrasts in sediment texture, bank cohesion, vegetation and flow conditions can produce spatially variable erosion and deposition. These regional characteristics are used in this study to interpret the observed morphology; lithology, grain size, bank stratigraphy and valley geometry were not measured at individual transects. Hydrologically, the reach is influenced by river discharge, seasonal water-level variation and, toward the downstream deltaic environment, tidal processes. Tidal effects are therefore treated as a plausible control on water-level and sediment dynamics, not as a variable directly measured for each Landsat acquisition.

The wider basin has experienced substantial anthropogenic pressure, including:

- Rapid population growth
- Urbanization
- Sand mining
- Dredging
- Navigation
- Oil and gas exploration
- Deforestation

These activities provide plausible local pressures on bank stability, but the present study did not independently measure dredging intensity, sand extraction volumes, vessel traffic, vegetation loss or oil infrastructure at each transect. Consequently, these factors are discussed as potential explanatory controls rather than as statistically tested causal predictors.

## 2.2 Research Design

The study adopted a multi-temporal geospatial design integrating Landsat imagery, the GIS and DSAS. The analysis was performed on the defined Bukuma–Tombia–Buguma reach. The four available observations are single-date images separated by approximately 20 years (1980–2000), 10 years (2000–2010) and 15 years (2010–2025). This sparse and uneven temporal sampling is explicitly recognized as a limitation: the study quantifies long-term endpoint displacement and fitted trends, but it does not represent continuous annual bank movement or seasonal variability.

The workflow comprised image acquisition and metadata screening, preprocessing and co-registration, bankline extraction, digitization and quality control, DSAS analysis, cross-sectional comparison, change classification, uncertainty screening, and interpretation as follows:

(a) Acquisition of the four Landsat scenes and recording of acquisition date, sensor, path/row, scene identifier, cloud cover and acquisition conditions.

(b) Radiometric/atmospheric preprocessing as appropriate to the product level, geometric checking, reprojection to a common coordinate reference system, and visual inspection for cloud, haze and missing data.

(c) Bankline extraction: the mapped feature was the instantaneous land-water boundary visible in each image, rather than the vegetation line or an assumed top-of-bank line. Water was delineated using multispectral contrast between water and adjacent land; the extracted boundary was then visually checked against the original multispectral composite. Because the Multispectral Scanner System (MSS), Enhanced Thematic Mapper Plus (ETM+) and Operational Land Imager-2 (OLI-2) have different spectral band configurations and spatial resolutions, the same physical interpretation rule—not identical band numbers—was applied across dates.

(d) Bankline digitization: banklines were digitized in a GIS from the preprocessed images, with manual editing used where mixed pixels, vegetation, shadows, cloud/haze or narrow channels obscured the boundary. The digitized



lines were inspected for topology, spikes, gaps and obvious sensor artefacts before the DSAS analysis. The digitization therefore represents a reproducible image-derived waterline, not a surveyed top-of-bank position.

(e) DSAS analysis: a baseline was constructed approximately parallel to the river and transects were generated approximately perpendicular to the banklines. Approximately 90 transects were used. The shoreline-change metrics calculated using DSAS included the EPR, LRR, WLR, NSM and SCE. The EPR, LRR and WLR were interpreted as rates ( $\text{m yr}^{-1}$ ); NSM and SCE were retained as distances (m).

(f) Cross-sectional analysis: representative transects at approximately 25%, 50% and 75% of the mapped reach were compared between the available years to assess lateral width adjustment and apparent bed-level change.

(g) Change classification: bankline segments were classified as accretion ( $>10$  m), stable ( $-10$  to  $+10$  m) or erosion ( $<-10$  m) using the signed displacement between mapped bank positions.

(h) Uncertainty and interpretation: results were screened against sensor resolution, cross-date registration uncertainty and the effects of different acquisition conditions; no causal attribution to individual human activities was attempted without independent measurements.

This workflow improves reproducibility, but the four-scene record remains too sparse to resolve short-term or seasonal dynamics. The reported long-term rates should therefore be interpreted as reach-scale estimates conditional on the image-derived waterline and its positional uncertainty.

### 2.3 Data Sources and Image Acquisition Metadata

Four Landsat observations were used. The methodological interpretation records the temporal gaps and the likely seasonal/tidal implications (Table 1).

Multitemporal Landsat imagery representing four observation epochs (1980, 2000, 2010 and 2025) was used to assess long-term changes in river-bank morphology. The datasets comprised Landsat 3 MSS imagery for 1980, Landsat 7 ETM+ imagery for 2000 and 2010, and Landsat 9 OLI-2 imagery for 2025. The selected scenes were acquired during the dry-season period to minimize cloud contamination and facilitate consistent inter-annual comparison. Scene metadata, including acquisition date, Worldwide Reference System (WRS) path/row, product identification and cloud-cover percentage, were recorded for each image. The 1980 Landsat 3 scene was referenced using the historical WRS-1 system, while the later Landsat scenes employed WRS-2 path/row notation.

**Table 1.** Landsat imagery and metadata used for the temporal analysis

| Epoch/Sensor         | Acquisition Date | WRS Path/Row   | Scene/Product Identifier                   | Cloud Cover | Metadata Status                                                                      |
|----------------------|------------------|----------------|--------------------------------------------|-------------|--------------------------------------------------------------------------------------|
| 1980—Landsat 3 MSS   | 15 January 1980  | WRS-1: 229/056 | LM3229056198 0015AAA00                     | 8%          | Historical Landsat 3 MSS scene; WRS-1 designation used for the early Landsat archive |
| 2000—Landsat 7 ETM+  | 04 March 2000    | WRS-2: 188/055 | LE7188055200 0064EDC00                     | 3%          | Landsat 7 ETM+ scene selected for the 2000 epoch                                     |
| 2010—Landsat 7 ETM+  | 13 December 2010 | WRS-2: 188/055 | LE71880552010 347EDC00                     | 5%          | Landsat 7 ETM+ scene; sensor designation corrected from Landsat 5 TM                 |
| 2025—Landsat 9 OLI-2 | 18 January 2025  | WRS-2: 188/055 | LC09_L1TP_188 055_20250118_20 250118_02_T1 | 4%          | Landsat 9 OLI-2 Collection 2 Level-1 scene selected for the 2025 epoch               |

Note: WRS = Worldwide Reference System; MSS = Multispectral Scanner System; ETM+ = Enhanced Thematic Mapper Plus; TM = Thematic Mapper; OLI-2 = Operational Land Imager-2.

### 2.4 Positional Accuracy, Uncertainty and Limitations

A formal co-registration root mean square error was not reported in the source manuscript and cannot be reconstructed without the original image-to-image control-point or DSAS uncertainty output. To avoid implying a measured accuracy that was not obtained, the revised analysis uses a conservative screening threshold based on image resolution. For the 60-m MSS image, a half-pixel positional uncertainty is 30 m; for the 30-m sensors, it is 15 m. Combining these independent half-pixel components in quadrature gives approximately 33.5 m as a cross-sensor screening uncertainty. Over 45 years, this corresponds to about  $0.75 \text{ m yr}^{-1}$  as an uncertainty-equivalent rate. The reported EPR ( $-1.63 \text{ m yr}^{-1}$ ), LRR ( $-1.37 \text{ m yr}^{-1}$ ) and WLR ( $-1.48 \text{ m yr}^{-1}$ ) are therefore larger than this screening rate, but this comparison is not a formal confidence test. Formal DSAS confidence intervals should be reported from the transect-level output in a final data-complete revision.

### 3 Results

#### 3.1 Overview of Riverbank Dynamics (1980–2025)

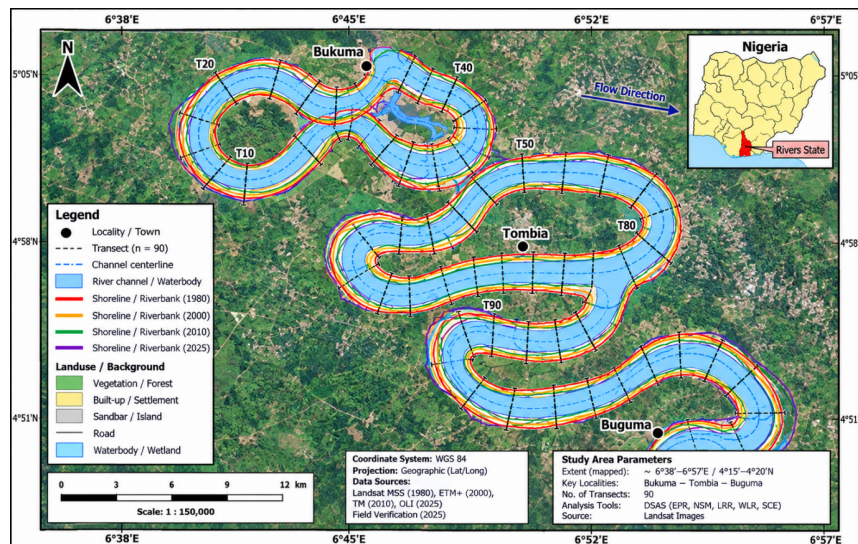
The multi-temporal analysis of the study area identified significant differences in the location of the riverbank during the 45-year period. By comparing the bank locations identified from the Landsat images in 1980, 2000, 2010, and 2025, it is possible to conclude that the dominant direction of the bank movement was related to erosion with a few exceptions where deposition took place. Using the resources of remote sensing and DSAS, it is possible to evaluate the trends in the movement of the riverbank using several statistical measures.

Based on the visual interpretation of the multi-temporal riverbank locations (Figure 2), it is possible to suggest that the channel shifted laterally during the analyzed period. In particular, the meandering loops of the river were the areas of extensive bank movement. It is possible to observe that the loops of the river that extended further than others experienced the most significant changes. On the straight dikes, the changes in the bank location were minimal. Similar to the natural meandering process of the rivers, the curvature of the channel appears to influence its lateral movement. Because the quantitative map ends at the defined Buguma downstream endpoint, the present dataset does not support a separate claim that erosion increased in the lower estuarine reaches beyond that endpoint. Tidal influence is discussed only as a regional process that may affect water levels and sediment transport within the deltaic setting.

Figure 2 compares the mapped banklines for 1980, 2000, 2010 and 2025 along the same Bukuma–Tombia–Buguma reach highlighted in Figure 1. The lines show substantial lateral adjustment in several meander bends, while relatively straight reaches exhibit smaller displacement. Approximately 90 transects were used for the DSAS analysis.

#### 3.2 Spatial Analysis of Historical Riverbank Positions

Figure 3 provides an enlarged view of the same Sombreiro River reach shown in Figure 2. The four banklines correspond consistently to 1980, 2000, 2010 and 2025. The observed offsets indicate lateral adjustment at several meander bends, while other portions show smaller displacement. Because the observations are discrete single-date images, the pattern should not be interpreted as proof of continuous annual migration. Furthermore, the shifts of the riverbank were not consistent across different areas. Some parts of the channel experienced erosion, while others remained stable. The observed changes can represent a combination of different effects, including the characteristics of the river flow, the local geology, vegetation, and human activity. Approximately 90 transects were established to analyze the movement of the riverbank by measuring their distance to the nearest bank at the time points identified in this study.



**Figure 3.** Map of the Sombreiro River study reach showing riverbank and shoreline positions (1980–2025), transects and key localities

Note: WGS = world geodetic system; MSS = Multispectral Scanner System; ETM+ = Enhanced Thematic Mapper Plus; TM = Thematic Mapper; OLI = Operational Land Imager; DSAS = Digital Shoreline Analysis System; EPR = end point rate; NSM = net shoreline movement; LRR = linear regression rate; WLR = weighted linear regression rate; SCE = shoreline change envelope.

### 3.3 Digital Shoreline Analysis System Statistics

The DSAS analysis yielded five statistical measures that describe distinct aspects of long-term riverbank migration (Table 2).

**Table 2.** Summary of DSAS statistics for the Bukuma–Tombia–Buguma reach (1980–2025)

| Statistic                             | Value and Unit           | Interpretation                                                                                                                                                                                          |
|---------------------------------------|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Net shoreline movement (NSM)          | −63.90 m*                | Reconstructed cumulative displacement from the source value ( $-1.42 \text{ m yr}^{-1} \times 45 \text{ years}$ ); the value should be verified against the original NSM output generated by DSAS.      |
| End point rate (EPR)                  | −1.63 m yr <sup>−1</sup> | Mean end-to-end retreat rate.                                                                                                                                                                           |
| Linear regression rate (LRR)          | −1.37 m yr <sup>−1</sup> | Negative fitted trend; based on four observations.                                                                                                                                                      |
| Weighted linear regression rate (WLR) | −1.48 m yr <sup>−1</sup> | Negative weighted trend; similarity to the EPR does not prove negligible registration error.                                                                                                            |
| Shoreline change envelope (SCE)       | 76.95 m*                 | Reconstructed maximum displacement envelope from the source value ( $1.71 \text{ m yr}^{-1} \times 45 \text{ years}$ ); the value should be verified against the original SCE output generated by DSAS. |

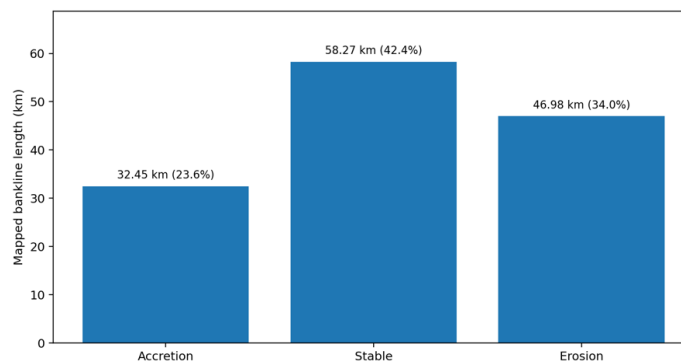
Note: DSAS = Digital Shoreline Analysis System; \* means NSM and SCE are distance statistics, not rates. The values shown are transparent reconstructions from rate-like values in the source manuscript and should be replaced by the original DSAS outputs if those outputs are available.

The EPR was  $-1.63 \text{ m yr}^{-1}$ , indicating net landward displacement of the mapped bankline over the 1980–2025 interval. Over 45 years, the magnitude corresponds to approximately 73.35 m of cumulative displacement if the rate is treated as representative of the full interval. The NSM is a distance statistic, not a rate. The source manuscript reported  $-1.42$  with rate units; if that value was produced by dividing the NSM distance generated by DSAS by 45 years, the corresponding displacement is  $-63.90 \text{ m}$ . Negative NSM denotes retreat in the adopted DSAS sign convention; it does not denote shoreline advance. This derived value must be checked against the original DSAS output. The LRR was  $-1.37 \text{ m yr}^{-1}$ , indicating a negative fitted trend in bank position through time. Because only four image dates were available, this trend should not be interpreted as evidence of uninterrupted annual erosion.

The WLR was  $-1.48 \text{ m yr}^{-1}$ . The WLR accounts for uncertainty weighting in the regression, but the similarity between WLR and EPR is not, by itself, evidence that co-registration error is negligible. The positional uncertainty must be evaluated independently. The SCE is a distance statistic and is non-negative. The source manuscript reported  $-1.71$  with rate units; if that value was derived by dividing a SCE distance by 45 years, the implied envelope is 76.95 m. This derived value is retained only as a transparent reconstruction and should be replaced by the original SCE output generated by DSAS if available. Taken together, the negative EPR, LRR and WLR values indicate a long-term tendency toward retreat within the mapped reach, while the NSM and SCE describe displacement magnitudes rather than rates.

### 3.4 Classification of Riverbank Change

Figure 4 summarizes the three classes of bankline displacement as linear bankline length: accretion, stable change and erosion. The figure is a statistical summary rather than a spatial map; the underlying values are reported in Table 3.



**Figure 4.** Classified bankline length for the Bukuma–Tombia–Buguma reach, 1980–2025

**Table 3.** Riverbank change statistics for the Bukuma–Tombia–Buguma reach (1980–2025)

| Category                   | Bankline Length (km) | Percentage (%) |
|----------------------------|----------------------|----------------|
| Accretion ( $>+10$ m)      | 32.45                | 23.6           |
| Stable ( $-10$ to $+10$ m) | 58.27                | 42.4           |
| Erosion ( $<-10$ m)        | 46.98                | 34.0           |
| Total                      | 137.70               | 100            |

Of the 137.70 km of mapped bankline, 58.27 km (42.4%) fell within the stable class. Erosion accounted for 46.98 km (34.0%) of the mapped bankline, indicating that approximately one-third of the analyzed bankline experienced displacement greater than the  $-10$  m erosion threshold. Accretion accounted for 32.45 km (23.6%) of the mapped bankline. These proportions describe the classified bankline and should not be interpreted as areal percentages of the river corridor.

### 3.5 Patterns of Erosion and Accretion

The mapped erosion tendency is most evident around strongly curved channel bends within the Bukuma–Tombia–Buguma reach shown in Figure 2 and Figure 3. In a meandering channel, the outer bank of a bend commonly experiences greater erosive stress, whereas the inner bank commonly favors deposition. The terminology and interpretation have been corrected in this study.

The concentration of erosion at strongly curved bends is consistent with lateral migration driven by spatial variation in flow velocity and bank resistance. Accretion was more evident along inner-bend and lower-energy portions of the mapped channel, while relatively straight and vegetated reaches showed smaller positional changes. The pattern of erosion and accretion was similar to the typical meander development. This pattern is consistent with established fluvial-meander behavior in which secondary circulation and velocity gradients redistribute sediment between outer-bank erosion zones and inner-bank depositional zones.

### 3.6 Digital Shoreline Analysis System Transect Framework and Cross-Sectional Limitation

Figure 5 shows the approximate distribution of the 90 DSAS transects along the same mapped reach. The supplied manuscript did not contain verified bathymetric or elevation-profile data for the 1980–2025 epochs. Therefore, the previous claims of apparent bed lowering, bed deepening, or vertical profile change have been removed rather than inferred from an unrelated figure. The transects are used in this study for bankline-displacement analysis only.

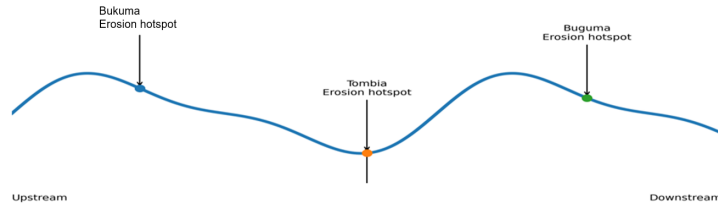
**Figure 5.** Digital Shoreline Analysis System transect framework along the mapped Bukuma–Tombia–Buguma reach (approximately 90 transects)

The middle portion of the reach contains a dense set of transects used to quantify lateral bankline displacement; no independent vertical bed change is inferred from these transects. The available evidence supports interpretation of lateral bank adjustment, but it does not support quantitative claims about vertical bed-level change. Transect density is maintained across upstream, middle and downstream portions of the mapped reach, allowing spatially distributed DSAS measurements rather than relying on a single cross-section.

### 3.7 Changes in the River Morphology Through Time

The mapped reach experienced a combination of bank erosion, bank accretion, meander migration and channel-width adjustment between 1980 and 2025 (Figure 3). This change was characterized by a combination of bank erosion, bank accretion, meander migration, and channel widening. The strongest changes occur in highly sinuous portions of the mapped reach. This spatial heterogeneity is consistent with fluvial meander dynamics, but the four-date record does not permit a claim of steady or continuous change between image dates. Below is a conceptual risk-context schematic for the mapped study reach (Figure 6). It illustrates potential planning exposure where settlements or agricultural land are close to retreating banks; it does not assign measured exposure or economic loss to individual locations.





**Figure 6.** Conceptual risk-context schematic of erosion hotspots and potential settlement or agricultural exposure along the mapped Bukuma–Tombia–Buguma reach

These findings provide a reach-specific baseline for future monitoring of the Sombreiro River. The results are interpreted in the context of established Niger Delta and tropical-delta research on riverbank erosion, sediment transport, channel adjustment and human intervention.

## 4 Discussion

### 4.1 Overview of Riverbank Dynamics

The revised results indicate a long-term tendency toward riverbank retreat within the defined Bukuma–Tombia–Buguma reach. The EPR, LRR and WLR are all negative, while the bankline classification shows 34.0% erosion, 23.6% accretion and 42.4% stable change. These results describe the mapped reach only; they do not establish that the entire Sombreiro River Basin or its lower estuarine reaches experienced the same rates. The four-date temporal record also means that the analysis should be interpreted as a multi-decadal comparison rather than a continuous erosion chronology.

The use of multiple DSAS metrics is appropriate because each metric describes a different component of shoreline movement. Hasan et al. [2] demonstrated the usefulness of combining shoreline-change metrics for riverbank assessment, while the DSAS framework provides standardized calculations for comparing shoreline positions [18]. The present revision follows that principle while correcting the units and interpretation of the NSM and SCE. Unlike the earlier version, the discussion does not treat negative NSM or SCE as rates and does not infer co-registration reliability from WLR alone.

### 4.2 Interpretation of Shoreline Change Statistics

The EPR of  $-1.63 \text{ m yr}^{-1}$  indicates a net retreat tendency over the 45-year interval. The LRR ( $-1.37 \text{ m yr}^{-1}$ ) and WLR ( $-1.48 \text{ m yr}^{-1}$ ) likewise indicate negative fitted trends. However, four observations cannot establish whether erosion occurred continuously between dates. The NSM and SCE are distance measures and should be interpreted as cumulative displacement and maximum displacement envelope, respectively. The original numerical presentation of the NSM and SCE was therefore corrected in Table 2.

### 4.3 Spatial Pattern of Riverbank Erosion

Spatially, erosion is concentrated around strongly curved bends of the mapped channel, especially around the Bukuma–Tombia–Buguma reach shown in Figure 2 and Figure 3. Such spatial heterogeneity is consistent with established fluvial meander dynamics, in which channel curvature and flow distribution contribute to alternating erosion and deposition [3, 19]. Similar spatially variable erosion and accretion patterns have been documented using geospatial analyses in other river systems [12–15]. The present data do not justify extending these hotspot observations beyond the mapped Figure 4 extent.

The observed alternation of erosion and accretion is therefore consistent with the morphology of meandering rivers. Hooke [19] described the role of channel curvature and velocity distribution in meander migration, while Knighton [3] provided the broader fluvial-process context. The agreement between these process concepts and the spatial pattern observed in the Sombreiro reach supports interpretation of the results as channel adjustment rather than uniform bank retreat.

### 4.4 Influence of Hydrological Processes

Hydrological conditions are likely to influence bank response through discharge, water-level fluctuations and, in deltaic reaches, tidal exchange. Abam [9] documented hydrological processes in the Niger Delta, while Abam and Omuso [10] examined river cross-sectional change in the region. Regional tropical-delta research also links sediment transport and geomorphological change to hydrological conditions and human intervention [4], while broader evidence shows that suspended-sediment regimes can change substantially in river deltas [11]. Because the present study did not have contemporaneous discharge or tide-gauge observations for each satellite acquisition, hydrological processes are interpreted as plausible controls rather than quantified predictors.

The hydrological interpretation is therefore constrained to process-based explanation. In particular, differences in water level between image acquisition dates may shift the instantaneous water–land boundary independently of long-term bank migration. This effect is especially important in tidal channels and is one reason the exact acquisition dates and tidal conditions should be retained with the raw imagery.

#### **4.5 Influence of Sediment Characteristics**

The alluvial character of the Niger Delta provides a plausible physical basis for bank instability. Abam and Omuso [10] showed that river cross-sectional change in the Niger Delta is influenced by bank materials, flow conditions and channel geometry. Studies of river-channel processes likewise emphasize the interaction among sediment transport, channel form and bank adjustment [3, 4, 20]. The present Landsat analysis, however, did not directly measure grain size, cohesion or stratigraphy at individual transects.

The sediment interpretation is therefore presented as contextual rather than as a measured causal relationship. Future field studies should combine remote sensing with bank-material sampling, hydraulic measurements and repeated cross-sectional surveys to test the mechanisms quantitatively.

#### **4.6 Role of Anthropogenic Activities**

Human activities such as dredging, sand extraction, navigation, vegetation clearance and settlement expansion may locally modify bank stability and sediment transport in the Niger Delta. Sand mining has been identified as an environmental pressure capable of altering river systems [5], while regional tropical-delta evidence demonstrates the geomorphological significance of human intervention [4]. Riverbank erosion can also generate socioeconomic consequences for exposed communities and assets [6, 17]. These activities are relevant contextual pressures, but the present dataset contains no time-matched records of their magnitude or spatial distribution. Accordingly, the revised manuscript avoids attributing individual erosion hotspots directly to specific anthropogenic activities.

This cautious interpretation is consistent with regional evidence that human intervention can alter deltaic sediment transport and channel morphology [4]. The management literature also emphasizes the need to identify erodible river corridors and integrate channel-process understanding into sustainable bank-erosion management [20, 21]. Future work should integrate dredging records, land-use change, vessel-traffic data and field observations before assigning quantitative causal effects to anthropogenic drivers.

#### **4.7 Interpretation of Cross-Sectional Riverbank Profile**

The three representative cross-sections provide independent visual evidence of lateral channel adjustment and apparent changes in profile geometry. However, because the profiles are derived from image-based bank positions rather than repeat field bathymetry, they cannot by themselves establish precise bed-level change. The strongest defensible conclusion is that the mapped channel widened or shifted laterally at selected transects, with spatially variable magnitude.

#### **4.8 Comparison with Previous Studies**

The findings are broadly comparable with previous Niger Delta and tropical-delta studies showing that riverbank and shoreline position can respond to discharge, sediment supply, tidal processes and human intervention. Regional Niger Delta studies have documented the importance of hydrological and cross-sectional controls on channel change [9, 10], while Restrepo et al. [4] demonstrated the importance of sediment transport and human intervention in a high-discharge tropical delta. These studies support a process-based interpretation but do not imply that their rates are directly transferable to the Sombreiro reach.

The comparison with recent remote-sensing studies supports the interpretation of spatially heterogeneous channel adjustment. Studies of the Ganga River and Bangladesh have demonstrated the utility of DSAS and multi-temporal imagery for quantifying erosion and accretion [12, 13], while the Barak River studies have documented temporal and spatial variability in channel erosion, accretion and morphology [14, 15]. Recent work using radar time series and autoregressive integrated moving average-based analysis has further illustrated the value of denser temporal observations for detecting river change [13, 16]. The socioeconomic dimension is also important, as recent research links riverbank erosion with livelihood impacts in exposed communities [17].

#### **4.9 Implication of the Findings**

The management implications concern the mapped reach and should be spatially targeted. Erosion-prone bank segments may threaten settlements, transport infrastructure, agricultural land, wetlands and other assets where development is close to the channel [6, 17]. Because the present study does not quantify exposure or economic loss, these implications are framed as planning concerns rather than measured impacts. The concept of identifying and managing erodible river corridors provides a useful framework for translating mapped bank instability into targeted management priorities [21].

The study recommends: (a) prioritizing monitoring and bank-protection planning at the mapped erosion hotspots; (b) retaining and restoring riparian vegetation where appropriate; (c) subjecting dredging and sand-extraction activities to site-specific environmental assessment [4, 5]; (d) avoiding new vulnerable development immediately adjacent to actively retreating banks; and (e) establishing repeat remote-sensing and field-survey programs using higher-resolution imagery, uncrewed aerial vehicles, global navigation satellite system and bathymetric measurements. These measures are consistent with the broader objective of managing erodible river corridors while accounting for channel processes and human exposure [20, 21].

#### 4.10 Strengths and Weaknesses of the Study

The principal strengths are the 45-year temporal span, integration of Landsat with GIS and DSAS, use of several complementary change statistics, and cross-sectional interpretation. The revised study also makes the spatial domain explicit and distinguishes rate statistics from distance statistics.

The principal limitations are equally important. Only four single-date images were available, with uneven temporal gaps; complete acquisition metadata were not preserved in the supplied manuscript; acquisition-season and tidal-stage effects could not be quantified; no empirical co-registration root mean square error or DSAS confidence intervals were available; and the earliest MSS imagery has a 60-m nominal spatial resolution. The conservative 33.5-m half-pixel screening uncertainty used in this study is not a substitute for a measured root mean square error. Future work should use a denser image time series, verified scene metadata, stable ground-control points, formal co-registration diagnostics, transect-level confidence intervals, and field validation.

### 5 Conclusion

This study assessed multi-decadal bankline change along the defined Bukuma–Tombia–Buguma reach of the Sombreiro River using four single-date Landsat observations from 1980, 2000, 2010 and 2025, GIS and DSAS. The quantitative reach covers approximately 6°38′–6°57′E and 4°15′–4°20′N, with an approximate 68.85-km channel length represented by 137.70 km of mapped bankline when both banks are included. The study therefore does not claim basin-wide erosion rates for reaches outside Figure 4. The classified bankline comprises 46.98 km (34.0%) of erosion, 32.45 km (23.6%) of accretion and 58.27 km (42.4%) of stable change. These are linear bankline proportions, not areas. The principal rate statistics are the EPR of  $-1.63 \text{ m yr}^{-1}$ , the LRR of  $-1.37 \text{ m yr}^{-1}$ , and the WLR rate of  $-1.48 \text{ m yr}^{-1}$ . The NSM and SCE are distance statistics; the values reconstructed from the original manuscript's incorrectly reported rates are approximately  $-63.90 \text{ m}$  and  $76.95 \text{ m}$ , respectively.

The distributed DSAS transects indicate that lateral bankline adjustment varies spatially along the mapped reach. Because verified bathymetric profiles were not available in the supplied manuscript, no quantitative vertical bed change is inferred. Bank response is plausibly influenced by channel curvature, hydrological variability, alluvial bank materials, vegetation and human pressure. However, the four-image dataset does not permit causal attribution of individual changes to tides, dredging, sand mining or other activities. Seasonal water-level and tidal-stage differences at acquisition may also affect the instantaneous waterline and therefore contribute to positional uncertainty. Remote sensing, GIS and DSAS provide a useful framework for long-term monitoring when the mapped domain, extraction rule, image metadata and uncertainty are explicitly reported. The present analysis should be treated as a baseline assessment of the defined reach. Future monitoring should use more frequent imagery, verified acquisition metadata, formal co-registration root mean square error, DSAS confidence intervals, field bank surveys and higher-resolution observations.

#### Author Contributions

Conceptualization, H.O. and T.P.A.; methodology, H.O. and T.P.A.; software, H.O.; validation, H.O., T.P.A., and M.B.A.; formal analysis, H.O.; investigation, H.O. and M.B.A.; resources, T.P.A.; data curation, H.O. and M.B.A.; writing—original draft preparation, H.O.; writing—review and editing, T.P.A. and M.B.A.; visualization, H.O.; supervision, T.P.A.; project administration, T.P.A. All authors have read and agreed to the published version of the manuscript.

#### Data Availability

The Landsat imagery used for this study is available at the United States Geological Survey (USGS) Earth Explorer database. The rest of the data generated for this study may be available upon request from the corresponding author.

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## Conflicts of Interest

The authors declare no conflicts of interest.

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